

SAI VIVEKANAND KUCHIMANCHI

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OBJECTIVE

Experienced Data Scientist, passionate about building robust and scalable solutions. Committed to pushing technological boundaries, delivering value, and staying at the forefront of AI advancements. Seeking collaborative opportunities with visionary teams to create transformative solutions that shape the future of AI.

CORE COMPETENCIES

- Worked on applying quantum computing techniques to real-world challenges in oil rig operations.
- Designed and developed a **Small Language Model (SLM)** for a **staffing / resourcing company**.
- Created a custom **Retrieval-Augmented Generation (RAG)** tool with :
 - Multi-stage architecture
 - Chunking and Embedding Metrics

Hands-on experience with all **5 wings of the GenAI Agent Ecosystem**:

- Intelligence / Brains (LLM's)
 - Experience with various **LLMs** (GPT-4, Claude, Mistral)
- Communications (11labs, Browser use, streamlit, Replit)
- Memory and Knowledge (Vector DBs (pinecone, weaviate, chroma db))
- Action systems (Agents > tasks > tools)
- Built and managed Agents :
 - Single and multi-agent systems.
 - Worked with several **external APIs** for tool integration.
- Orchestration and Monitoring :
 - MCP servers
 - **Agent Development Kits (ADKs)**: OpenAI ADK, Google ADK
 - Automation tools (n8n)
 - **Frameworks**: LangChain, LangGraph (for multistep orchestration)
 - **Multi-agent platforms**: CrewAI, Playwright, Agno, Lovable

Worked closely with Manufacturing Industry use-cases.

Industry 4.0 , OEM, Digital Twin, IIoT, Visual Inspection, Predictive Maintenance

Used AI based solutions in solving Manufacturing problems.

WORK EXPERIENCE - 14Yrs

- DATA SCIENTIST, HELIX BEAT (AUG 2024 - Till Date)
- Sr. DATA SCIENTIST, ADYNT (AUG 2022 - AUG 2024)
- Sr. DATA SCIENTIST, DATAMATICS (MAY 2022 - AUG 2022)
- DATA SCIENTIST, TECH M (SEPT 2017 - MAY 2022)

- RPA SOLUTION ARCHITECT, TCS (JUN 2014 - MAY 2017)
- LEAD SAP BO BI, ACCENTURE (OCT 2010 - JUN 2014)

MAJOR PROJECTS

• GEN AI - RATE ANALYSIS (JUN 2024 - FEB 2025)

- Developed a solution for plastic injection Molding machine for Motherson Automotive Technologies and Engineering (MATE), chennai. Using computer vision and ORB (Oriented FAST and Rotated BRIEF) feature descriptor, compare live feed image and reference image to instruct the molding machine to start the next cycle or to terminate next cycle and terminate the production.

Relevant Activities :

- * Document conversion, Llama-parser for document chunking. Document embedding (Azure embedding, Hypo embedding). Model - "Text-embedding ada-002". Python.
- * Document indexing (Langchain), FAISS vector DB, Azure Blob storage.
- * Angular, Web API, SQL Server, Langchain OPENAI (LLM).
- * Azure OPEN AI API (GPT4). FAISS Similarity Search, Hyde Retriever.
- * Train the model on various documents. Check accuracy with labels.
- * Web farmework using Angular.

• CV2 for INJECTION MOLDING MACHINE - (MATE) (FEB 2024 - MAY 2024)

- Developed a streamlined solution for rate case filings that optimize document submission, querying, and analytics. Reduce regulatory team workload through automation and enhance compliance. Enabled utility companies to adapt to market changes and drive sustainable growth with improved insights and operational efficiency.

Relevant Activities :

- * Coordinated with MATE engineers to understand machine workflow and automation checkpoints. Defined operational parameters for acceptable part quality using visual cues.
- * Captured and labeled reference images of correctly molded parts for training and validation. Recorded live production feed for comparison using camera.
- * Implemented ORB (Oriented FAST and Rotated BRIEF) feature descriptors for key-point extraction. Tuned image preprocessing pipeline (grayscale conversion, thresholding, denoising, etc.).
- * Developed OpenCV-based image matching logic to compare live feed with reference image. Implemented logic to compute homography and match confidence scores.
- * Performed daily tests on live production units to ensure detection reliability under varying lighting and part conditions. Logged and debugged false positives/negatives in visual comparison output.
- * Fine-tuned ORB parameters for speed and accuracy trade-off in embedded hardware. Optimized pipeline latency to keep cycle time within acceptable bounds (milliseconds delay).
- * Maintained daily logs of testing results, false match cases, and improvements. Created user documentation for operators to understand system usage and behavior.
- * Participated in daily stand-up or syncs with project team and MATE engineers. Reviewed integration feedback from the shop floor and revised logic accordingly.

• RARE ADENOID CANCER CLASSIFICATION using SVM (Sep 2023 - Feb 2024)

- Biomedical Data | diagnose adenoid cancer in early stages | Visualization: pairplots, countplot, scatterplot, heatmap| Confusion Matrix | Model improvement: Feature scaling, Optimize C and Gamma parameters using Grid Search | Ensemble methods.

Daily Activities :

- * Collect new data samples | ensure data quality | update dataset. Continuously tune model hyper-parameters to adapt to changes in the data distribution and improve performance.
- * Perform grid search or random search to find the optimal hyperparameters.
- * Performing cross-validation to ensure the model's robustness.
- * Managing the inference pipeline, ensuring smooth updates and minimal downtime.
- * CI/CD pipeline using Jenkins. Developed using Python.

• **ANOMALY DETECTION using AUTOENCODER** (May 2023 - Aug 2023)

- Financial Data | Torch tensor | Encoder - Decoder | MSE Reconstruction loss | Adam Optimizer.

Daily Activities :

- * Collect New Data |Retrain the Model |Hyper-parameter Tuning
- * Try iForest | Regularly evaluating the model using various metrics.
- * Model Validation | Run Anomaly Detection in Training pipeline
- * Evaluate Anomalies | Monitor Model Performance , Log and Analyze Anomalies.
- * Remove highly co-related features. Deploy the trained model to a production environment for real-time anomaly detection.
- * Update Thresholds , Monitor Data drift, Model drift.
- * Pytorch framework, CI/CD pipeline using Jenkins. Developed using Python and its libraries.

• **OPTUS - CUSTOMER ANALYTICS** (Feb 2018 - Aug 2018)

- CUSTOMER SEGMENTATION, SENTIMENT ANALYSIS, CHURN PREDICTION
- Telecom Data | RFM score for Customer segmentation | VADER for Sentiment analysis | Random Forest classifier for Churn prediction

Daily Activities :

- * Applied K-means, hierarchical clustering, RFM analysis. Tried Logistic Regression, or Gradient Boosting algos. Python, angular and Azure.
- * Deal with stopwords, punctuation. Tokenize, lemmatize to prepare text.
- * Visualized sentiment trends for customer segments. Provide actionable recommendations
- * Alerted stakeholders on emerging sentiment trends. Churn patterns, churn rate, churn metrics.
- * Engineered features such as customer tenure, usage metrics, and customer interactions.

• **CMS - BRIEF BANK - LAW LIBRARY** (Aug 2022 - Jan 2023)

- Apply elastic search on Law library, to get relevant case files from millions of case files.

Daily Activities :

- * Create html page. Render it using flask. Plaintiff submits keywords.
- * Data is scraped, opens US Law Library, applies elastic search on all the 1.5 billion court verdicts. Python and AWS.
- * Pytesseract to convert the handwritten file images to text.

* Using a get request json file. Provide download button using href localhost 8080 html code .

TECHNICAL SKILLS

- * Programming Languages : Python, VBE |Databases : MS SQL Server, Mysql
- * Angular, CI / CD pipelines
- * Visualization : Tableau, Power BI, Excel |Cloud Platforms : AWS, Azure
- * ML : Data IKU, Databricks |Frameworks : sklearn, automl, tensorflow, keras, pytorch
- * Domains : Finance, Biomedical, Marketing (Customer Analytics), Recommender systems
- * Technologies : NLP, GEN AI (LLM)

ACADEMIC DETAILS

TITLE	UNIVERSITY	INSTITUTE	YEAR(S)
Bachelor’s Degree - B.Sc. HMCS	Madurai Kamaraj University	Kodaikanal Christian College	1996 - 1999
Data Science, Machine Learning	Scaler Academy	Scaler Academy	2022 - 2024